

THAKUR ADITYA SINGH

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📍 Hyderabad, India

Work Experience

LTIMindtree

Dec 18, 2024 – Present

Software Engineer

Hyderabad, India

- Working as a Full-Stack Software Engineer developing enterprise-grade applications using Java, Spring Boot, Angular, React, Vite, Node.js, and Express, delivering scalable and maintainable web platforms.
- Designed and implemented RESTful APIs using Spring Boot and Express.js, enabling seamless communication between frontend and backend services.
- Developed responsive and high-performance user interfaces using React, Angular, HTML5, CSS3, and modern JavaScript, improving application usability and responsiveness.
- Built backend microservices using Node.js and Express framework, integrating PostgreSQL databases to support efficient data processing and high availability.
- Optimized database queries and schema design in PostgreSQL and MySQL to improve application performance and reduce data retrieval latency.
- Implemented CI/CD pipelines using Jenkins to automate build, testing, and deployment processes, reducing manual deployment effort and improving delivery speed.
- Managed version control and collaborative development using Git, actively contributing to code reviews and maintaining clean, maintainable codebases.
- Participated in Agile development cycles including sprint planning, backlog grooming, daily stand-ups, and retrospectives to deliver features efficiently.
- Integrated secure authentication and authorization mechanisms using modern API security practices to ensure safe access to application resources.
- Improved system performance and scalability by optimizing backend services and implementing efficient API design patterns.
- Collaborated with cross-functional teams including QA engineers, product managers, and designers to deliver production-ready features aligned with business requirements.
- Troubleshoot and resolved production issues, ensuring system reliability and minimizing downtime through efficient debugging and monitoring.

Worldlynk Limited (Remote)

Feb 2024 – Oct 2024

Freelance Full Stack Developer

London, United Kingdom

- Architected and developed a web platform for international students aspiring to study in the UK, combining React.js, Redux, Tailwind CSS, and JavaScript to deliver a seamless user experience and boost engagement by 25%.
- Designed a scalable backend using Firebase services (Firestore with optimized indexes, structured subcollections, and secure rules) and microservices in Express/Node.js deployed on Netlify, reducing infrastructure overhead.
- Implemented an end-to-end consultancy workflow enabling consultants to manage and track student applications efficiently, improving operational transparency and service delivery.
- Integrated Stripe as a secure payment gateway for event bookings, restaurant reservations, and accommodation services, ensuring compliance with industry standards.
- Built a real-time messaging system using Firebase Realtime Database, enabling instant communication between students and consultants.
- Designed and executed a full ETL pipeline for ingestion, transformation, and storage of 1,000+ structured university data points, enhancing decision-making resources.
- Optimized application performance by reducing Firestore read requests and improving load times by 30%, significantly enhancing efficiency and user satisfaction.
- Developed and launched the “Reliv” social feature, enabling users to share experiences and increasing platform engagement metrics by 25%.

Education

Anurag University

Jun 2020 – Jun 2024

B.Tech in Artificial Intelligence (GPA: 8.07)

Hyderabad, India

Technical Skills

Languages & Core: Python, SQL, JavaScript (ES6+), TypeScript, C++, C#, Java.

Frameworks & Architecture: .NET Core / .NET Framework, ASP.NET MVC, Spring Boot, Express.js, RESTful APIs, Microservices, Clean Architecture, SOLID Principles.

Frontend Technologies: React.js, Angular (16–18), Redux, Tailwind CSS, HTML5, CSS3, Responsive Design.

Cloud & DevOps: Microsoft Azure, AWS (EC2, S3, Lambda), Docker, Kubernetes, Azure DevOps, GitHub Actions, CI/CD Pipelines.

Databases & Storage: SQL Server, PostgreSQL, MySQL, MongoDB, Firebase Firestore, Redis (caching, pub/sub).

Security & Identity: OAuth2.0, JWT, LDAP, Azure Active Directory, Role-Based Access Control (RBAC).

Projects

DevTrack: Personal Productivity & Monitoring Suite

2025

React.js, Node.js, PostgreSQL, Docker, Kubernetes, Grafana

- Developed a full-stack productivity suite integrating a React.js frontend with a Node.js backend, enabling real-time analytics and performance tracking.
- Implemented secure authentication and role-based access control using JWT, ensuring data privacy and controlled access.
- Designed ETL pipelines to process daily activity logs into PostgreSQL, supporting advanced queries and actionable insights.
- Integrated Grafana dashboards for live monitoring of coding hours, error rates, and productivity metrics, enhancing developer observability.
- Containerized the application with Docker and deployed on Kubernetes, demonstrating end-to-end DevOps proficiency.

VisionCare: Deep Learning Medical Imaging System

2024

Python, TensorFlow, Keras, OpenCV, AWS

- Built a deep learning model using CNNs to detect early signs of diabetic retinopathy from retinal images, achieving 92% classification accuracy.
- Applied transfer learning with pre-trained models (ResNet, EfficientNet) to improve accuracy and reduce training time.
- Implemented preprocessing pipelines with OpenCV for image normalization, augmentation, and noise reduction.
- Deployed the trained model on AWS EC2 with RESTful APIs, enabling scalable access for healthcare providers.
- Integrated monitoring dashboards to track inference latency and model performance in production.

ConversAI: Multimodal AI Chatbot and Image Intelligence System

2024

Python, OpenAI API, HuggingFace, LangChain, PyTorch, OpenCV, FastAPI, React, Docker, AWS

- Developed a multimodal AI assistant capable of human-like conversational interaction using Large Language Models (LLMs) integrated through OpenAI APIs and LangChain orchestration.
- Implemented semantic search using embedding models and FAISS vector databases to enable contextual question answering and knowledge retrieval.
- Built an automated chatbot capable of natural dialogue, contextual memory handling, and dynamic response generation for interactive user assistance.
- Designed an image classification pipeline using deep learning models with TensorFlow and PyTorch to automatically categorize and analyze uploaded images.
- Implemented preprocessing and feature extraction pipelines using OpenCV to enhance image quality and improve model prediction accuracy.
- Developed REST APIs with FastAPI and deployed the system using Docker containers on AWS for scalable inference and real-time interaction.

Learning Destiny: Scalable Education Platform

2023

Angular, TypeScript, Firebase, WebSockets

- Built a scalable, real-time education platform with Angular and Firebase, supporting thousands of concurrent users.
- Implemented cross-cloud data synchronization and granular role-based access control (RBAC) for secure collaboration.
- Optimized Firebase queries and WebSocket connections, reducing read costs by 20% and improving responsiveness.

Certifications & Leadership

- **VMware IT Academy – Modern Applications, Containers & Kubernetes:** Completed a 12-hour technical certification covering containerized application development, Kubernetes orchestration, and modern cloud-native deployment practices.
- **Technical Mentor at LearningDestiny Academy:** Mentored 50+ students in Data Structures & Algorithms and full-stack development, helping learners strengthen problem-solving skills and practical software engineering knowledge.
- **University Jury Member / Technical Evaluator:** Invited as a jury member to evaluate student software engineering projects and technical presentations, providing feedback on implementation quality, architecture, and problem-solving approaches.